

gamlssPosthoc 0.3.0

Marginal, distributional, and graphical post-hoc inference for GAMLSS

Decision flow

- 1. Declare the estimand: parameter / mean / variance / quantile / $P(Y=0)$ / custom.
- 2. Declare target population: observed / balanced / reference.
- 3. Choose scientific comparison: difference / ratio / log ratio / percent change.
- 4. Run `gamlss_posthoc_plan()` before complex inference.
- 5. Choose uncertainty: delta / simulation / refit bootstrap / none.
- 6. For figures, use `gamlss_plot_data()` first and plot second.

Core functions

- `gamlss_posthoc()` - standardized estimates and contrasts
- `gamlss_posthoc_plan()` - engine/uncertainty diagnostic
- `gamlss_distribution_summary()` - mean, variance, quantiles, $P(Y=0)$
- `gamlss_trend()` - curve, derivative, turning points, optimum

Scientific plots

- `plot_gamlss_parameters()` - all detected distributional parameters
- `plot_gamlss_estimand()` - mean/variance/quantile/ $P(Y=0)$
- `plot_gamlss_contrasts()` - estimation/forest or ggdist display
- `plot_gamlss_distribution()` - density/CDF/survival
- `plot_gamlss_quantiles()` - lines or fan chart
- `plot_gamlss_zero_adjusted()` - zero + positive + marginal components
- `plot_gamlss_trend()` / `derivative()` / `optimum()`
- `plot_gamlss_surface()` - 2D heatmap/contours
- `plot_gamlss_fit()` - observations + predictive quantiles
- `plot_gamlss_diagnostics()` - residuals, QQ, worm-style, PIT, rootogram
- `plot_gamlss_compare()` - criteria or estimand curves

Key formulas

- Standardization: $T_g = \sum_i w_i T(F_{ig}) / \sum_i w_i$
- Zero-adjusted mean: $E(Y) = (1-h) E(Y+)$
- Zero-adjusted variance: $Var(Y) = (1-h)Var(Y+) + h(1-h)E(Y+)^2$
- Difference: $A-B$; Ratio: A/B ; Log-ratio: $\log(A/B)$; Percent change: $100(A/B-1)$

Quick syntax

Goal	Minimal syntax
Plan	<code>gamlss_posthoc_plan(fit, estimand="mean")</code>
Means	<code>gamlss_posthoc(fit, "trt", estimand="mean")</code>
% change	<code>gamlss_posthoc(..., contrast="reference", comparison="percent_change")</code>
Parameters	<code>plot_gamlss_parameters(fit, "dose", parameters="all")</code>
Distribution	<code>plot_gamlss_distribution(fit, "trt", type="density")</code>
Quantile fan	<code>plot_gamlss_quantiles(fit, "dose", style="fan")</code>
Trend	<code>plot_gamlss_trend(fit, "dose")</code>
Derivative	<code>plot_gamlss_derivative(fit, "dose", order=1)</code>
Optimum	<code>plot_gamlss_optimum(fit, "dose")</code>
Surface	<code>plot_gamlss_surface(fit, "dose", "salinity")</code>
Diagnostics	<code>plot_gamlss_diagnostics(fit, "all")</code>
Tidy	<code>generics::tidy(result)</code>
Word	<code>export_to_word(result, "result.docx")</code>
Report	<code>generate_report(result, "html", "report.html")</code>

Interpretation rules

- Do not call μ a mean unless the family parametrization guarantees it.
- For zero-adjusted models, distinguish $P(Y=0)$, $E(Y|Y>0)$, and marginal $E(Y)$.
- Use bootstrap for derived quantities when a joint covariance interface is not trustworthy.
- A mathematical maximum is not automatically an agronomic/economic recommended dose.
- Compact letters are display aids, not evidence of equivalence.

References: Rigby & Stasinopoulos (2005), doi:10.1111/j.1467-9876.2005.00510.x; Arel-Bundock et al. (2024), doi:10.18637/jss.v111.i09; Kay (2024), doi:10.1109/TVCG.2023.3327195; Dunn & Smyth (1996), doi:10.1080/10618600.1996.10474708.